MIL-D-11933B
1 FEBRUARY 1955
SUPERSEDING
MIL-D-11933A (SigC)
2 FEBRUARY 1954

MILITARY SPECIFICATION

DUMMY, LOAD, ELECTRICAL DA-64( )/UP

This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of resistive network designated as Dummy, Load, Electrical DA-64( )/UP. This test gear shall be capable of dissipating radar-modulator pulse energy and shall also serve as a voltage divider for test purposes. (See 6.3.)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 —Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 —Boxes, Wood, Wirebound (For Domestic Shipment.)
PPP-B-601 —Boxes, Wood-Cleated-Plywood.

MILITARY

JAN-P-106 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed (for Weight of Contents not in Excess of 1,000 Pounds.)
MIL-B-107 —Boxes, Wood, Wirebound (Overseas Type.)
JAN-P-108 —Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior.
MIL-P-116 —Preservation, Methods of.
JAN-P-117 —Packaging and Packing for Overseas Shipment—Bags, Interior Packaging.
JAN-P-127 —Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water Resistant.
MIL-M-13231 —Marking of Electronic items.

U. S. ARMY

19-10 —Filler, Interior Packing, Fiberboard, Corrugated, Single and Double Faced, For General Packing Purposes.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 —Marking of Shipments.
MIL-STD-130 —Identification Marking of U. S. Military Property.
MIL-STD-252 — Wired Equipment, Classification of Visual and Mechanical Defects for.

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DRAWINGS

SIGNAL CORPS

SC-D-15914 — Moisture Resistance Test Cycle For Ground Signal Equipment.
SC-D-19999 — Standard Extreme-Temperature Cycle.
SC-DL-87564 —Dummy, Load, Electrical DA-64( )/UP, Drawing and Data List.

(Copies of specifications, standards, and drawings required by contractors in connection with specified procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications. — The following document, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification.

ASSOCIATION OF AMERICAN RAILROADS
Uniform Freight Classification-Ratings, Rules and Regulations.

(Application for copies should be addressed to the Association of American Railroads, Chicago, Illinois.)

3. REQUIREMENTS

3.1 Construction.—Dummy Load, Electrical DA-64( )/UP shall be constructed in accordance with Drawing and Data List SC-DL-87564. In addition, the equipment shall meet the requirements of this specification.

3.2 Preproduction samples.

3.2.1 *Quantities.* — After award of contract, two preproduction samples shall be furnished for preproduction inspection and approval.

3.2.2 *Compliance with specified requirements.*

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.2.2.3 for exception.)

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.2.5), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.2.3 When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.2.3 *Test data.*—The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like.)

3.2.4 *Preproduction inspection.*—Preproduction samples may be subjected by the Government to the group A inspection specified in Table II and to the preproduction inspection specified in Table I. Unlisted nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements.

TABLE I. *Preproduction inspection*

Inspection	Requirement paragraph	Test paragraph
Temperature	3.11	4.2
Moisture-resistance	3.11	4.3
Immersion	3.11	4.4

3.2.5 *Reference standards.* — After preproduction samples have been approved and returned to the contractor's plant, they shall

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be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any difference of opinion regarding interpretation of requirements.

3.3 Technical literature, and running spare parts. — Technical literature and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.4 Plastic material. — Where not machined, plastic material shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the kind of material used and the intended application.

3.5 Soldering.

3.5.1 *Flux and cleaning agents.*—No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin or rosin and alcohol, or rosin and turpentine.

3.5.2 *Process.*—There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. The soldering shall be so accomplished that no rosin remains on the joint or in the proximity of the joint. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment, flux being spread over a larger area, or resulting damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.6 Cleaning.—Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.7 Welding.—Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. Wherever possible, the welded joint shall be such that grinding on the finished weld will be unnecessary.

3.7.1 *Cleaning.* — Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned, to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.7.2 *Process.*—Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks,

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and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum or magnesium.

3.8 Interchangeability. — Corresponding components and replaceable subassemblies and parts on contract shall be physically and functionally interchangeable as units without modification thereof or of other items with which the units are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. Whether subassemblies and parts are to be considered replaceable shall be as determined by the contracting officer.

3.9 Identification of equipment. — The equipment shall be marked for identification in accordance with Standard MIL–STD–130.

3.9.1 *Marking of electronic items.*—Marking of electronic items shall be in accordance with Specification MIL–M–13231.

3.10 Electrical details.

3.10.1 *Resistance.* — The terminating resistance shall be 50 ohm ±5 percent.

3.10.2 *Reactance.* — The reactance of the dummy load shall not exceed 2 ohms at any frequency up to 3 megacycles per second.

3.10.3 *Dissipation.* — The dummy load shall be capable of dissipating at least 1,200 watts average power.

3.10.4 *Voltage divider.*—The dummy load shall be capable of acting as a voltage divider with a ratio of 100:1 for all voltages up to 10,000 volts peak. The dividing ratio shall be accurate to within 10 percent.

3.11 Service conditions. — The equipment shall meet the following service conditions:

3.11.1 *Equipment operating.*—The equipment shall meet the requirements of this specification under any probable combination of the following:

(a) *Operation.* Continuous operation in any specified manner, with no more than normal maintenance and replacement of parts. Compliance shall be determined by examination of the preproduction sample. (See 3.2).

(b) *Temperature.* Any ambient temperature in the range of –40° to +150°F. (The 150°F., temperature includes effect of sun load). (See 4.2).

(c) *Elevation.* Any elevation up to 10,000 feet above sea level.

(d) *Relative humidity.* Any relative humidity up to 98 percent.

3.11.2 *Equipment nonoperating.* — The equipment shall meet the requirements of 3.11.1 after subjection to any one or more of the following nonoperating conditions:

(a) *Temperature.* Continuous exposure for 72 hours at 160°F., and 72 hours at –80°F., at any orientation.

(b) *Elevation.* Any elevation up to 50,000 feet above sea level.

(c) *Relative humidity.* Relative humidity up to 100 percent, including condensation caused by temperature changes. (See 4.3)

(d) *Orientation.* Indefinite storage in any orientation.

(e) *Immersion.* See 4.4.

3.12 Workmanship.—The equipment shall be manufactured and assembled in a thoroughly workmanlike manner in accordance with the applicable portions of paragraphs 3.4 thru 3.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection, general.—Government inspection shall consist of group A inspection as specified in 4.1.1.

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4.1.1 *Group A inspection.*—Group A inspection shall be in accordance with Table II and Standard MIL–STD–105.

TABLE II. *Government group A inspection*

Inspection	Requirement paragraph	Test paragraph	AQL	Control
Visual and mechanical, major.	-------	4.5	1% or dphu	25 dphu
Visual and mechanical, minor.	-------	4.5	4% or dphu	25 dphu
Resistance	3.10.1	4.6	1% for subgroup	--------
Reactance	3.10.2	4.7		--------
Dissipation	3.10.3	4.8		--------
Voltage divider.	3.10.4	4.9		--------

4.1.2 *Disposition of nonconforming product.*—When defective sample units or rejected lots are resubmitted for acceptance, such items shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the items are resubmitted and shall become the property of the Government.

4.2 Temperature test for equipment.—The equipment shall be subjected to the temperature cycle shown on Drawing SC–D–19999. Measurements shall be taken as follows:

Paragraph	*Step*	*Measurement*	*Test results*
4.6	1,3,5,8,10	Resistance	Full specification performance
4.7	1,3,5,8,10	Reactance	Full specification performance

4.3 Moisture-resistance test for equipment.

4.3.1 *Test conditions.*

(a) Do not move components from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the components. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.3.2 *Test procedure.*—The equipment will be tested as follows:

(a) Dry at 130° ±5°F. for 24 hours.

(b) Condition at 77° ±5°F. and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.3.3.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Drawing SC–D–15914. The measurements specified in 4.3.3 shall be made during the periods shown on the drawing and shall comply with 4.3.3.

(e) After cycling has been completed, condition the components for 24 hours at 77° ± 5° F. and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet full specified performance for those measurements specified in 4.3.3.

4.3.3 *Performance.*—Measurements made during test, and specified performance during cycling, shall be as follows:

Paragraph	*Measurement*	*Performance during cycling*
4.6	Resistance	Full specification performance
4.7	Reactance	Full specification performance

4.3.4 *Failure.*—If the equipment fails to meet the performance specified in 4.3.3 during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning as specified in 4.3.2(e), it does not pass the test.

4.4 Immersion test.—The equipment, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment

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shall be approximately 40° F. above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water at approximately its initial temperature, or the temperature of the water shall be maintained constant by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened there shall be no evidence of leakage.

4.5 Visual and mechanical inspection. — Parts and equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.6 Resistance test.—The equipment shall be tested to determine compliance with the requirements of 3.10.1.

4.7 Reactance test.—The equipment shall be tested to determine compliance with the requirements of 3.10.2.

4.8 Dissipation test.—The equipment shall be tested to determine compliance with the requirements of 3.10.3.

4.9 Voltage divider test.—The equipment shall be tested to determine compliance with the requirements of 3.10.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 *For immediate use at first domestic receiving activity.*—Dummy, Load, Electrical DA-64( )/UP shall be preserved and packaged to afford protection against deterioration and damage during shipment.

5.1.2 *For entry into the supply system.*—Preserve and package Dummy, Load, Electrical DA-64( )/UP in accordance with procedures specified for the designated ' methods as prescribed in Specification MIL-P-116 and further described in the following paragraphs.

5.1.2.1 *Technical literature.* — Package each technical literature method IC-3, as follows: Inclose each technical literature within a close-fitting class b bag, conforming to Specification JAN-P-117. Properly seal closure.

5.1.2.2 *Dummy, Load, Electrical DA-64 ( )/UP, (together with cable assemblies and adapters).*—Package each complete equipment method IC-5, as follows: Coil cables and place in space provided. Replace covers of component and secure with the fastenings provided. Cushion each equipment on all surfaces with cells and pads, fabricated of corrugated fiberboard (A or C flute acceptable) conforming to Specification 19-10 designed to absorb the shock of impact normally encountered in handling and transit. Place the cushioned item within a close-fitting suitable style and grade fiberboard box, conforming to the requirements of Specification JAN-P-108. Seal entire closure, corners and joints with water-resistant pressure-sensitive tape, conforming to Specification JAN-P-127.

5.2 Packing.

5 2.1 *For immediate use at first domestic receiving activity.*—Dummy, Load, Electrical DA-64( )/UP shall be so packed as to insure arrival at destination in satisfactory condition and in containers complying with Uniform Freight Classification Rules at lowest rate.

5.2.2 *For domestic shipment and indefinite storage.*—Place a quantity of Dummy, Load, Electrical DA-64( )/UP in wood-cleated-plywood, nailed wood, or wirebound, conforming to Specification PPP-B-601, NN-B-621, or NN-B-631 respectively. The gross weight shall not exceed approximately 200 pounds. Closure shall be made in accordance with the applicable box specification.

5.2.3 *For overseas shipment.* — Place a quantity of Dummy, Load, Electrical DA-64 ( )/UP to be packed, in wood-cleated-plywood, nailed wood, or wirebound boxes, con-

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forming to Specification PPP–B–601 (Overseas Type), JAN–P–106, or MIL–B–107 respectively. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping container, as specified in the appendix of the applicable box specification.

5.2.3.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification *only* for direct shipment to ports.

5.3 **Marking for shipment.**—Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.**—The dummy load is intended for use as a load on the radar modulator in lieu of the magnetron and for use as a voltage divider to allow visual analysis of the modulator output.

6.2 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether domestic or overseas shipment is required. (See 5.2.)

(c) Marking and shipping of samples.

(d) Place of final inspection.

(e) Technical literature required. (See 3.3.)

(f) Quantity of running spare parts required. (See 3.3.)

6.3 **Nomenclature.** — The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example: DA–64W/UP. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information. (See 1.1.)

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation, whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Signal Corps
Navy—Marine Corps
Air Force

Other Interest:

Army—O.

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